

$$Q1.2: \text{ov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))((a + by_i) - m(a + bY));$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m((a + by_i) - (a + bY))) =$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))((a_i + by_i) - (a + b * m(Y))) =$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(by_i - b * m(Y)) =$$

$$\frac{1}{N} \sum_{i=1}^N b(x_i - m(X))(y_i - m(Y)) = b \times \text{cov}(X, Y)$$

$$Q1.3: \text{cov}(a + bX, a + bX) = \frac{1}{N} \sum_{i=1}^N ((a + b X_i) - m(a + bX))((a + b x_i) - m(a + bX)) =$$

$$\frac{1}{N} \sum_{i=1}^N (bx - b * m(X))(bx - b * m(X)) = \frac{1}{N} \sum_{i=1}^N b(x - m(X))b(x - m(X)) =$$

$$\frac{1}{N} \sum_{i=1}^N b^2(x - m(X))(x - m(X)) = b^2 \text{cov}(X, X); \text{ without } b^2, \text{ you would have:}$$

$$\frac{1}{N} \sum_{i=1}^N (x - m(X))(x - m(X)) = \frac{1}{N} \sum_{i=1}^N (x - m(X))^2 = s^2$$

Q1.4: No, that property only applies if the transformation of the mean is linear, which was the case with the constants in front of the random variables. If a variable is nonlinear, then one cannot preserve the averages since the space between different values. For example, if there is a function $g(x) = x^2$, and you have the set $X = [1, 3]$, $m(g(x)) = \frac{1+(3)^2}{2} = 5$, whereas $g(m(X)) = g(\frac{1+3}{2}) = g(2) = 4$, and $5 \neq 4$.